SIGMIDI Foot Controller v0.1 — electrical diagrams

The Nano ESP32 assembly has a unified A0 schematic, eight detailed A3 sheets, browser preview, terminal CSV and machine-readable netlist. Purchased boards stop at external pins. These are assembly wiring drawings, not PCB fabrication files. Bench verification is required.

U1 is Arduino Nano ESP32 ABX00083, component-side top view with USB-C up. JP1 is left, pads 1–15 top-down; JP2 is right, pads 15–1 top-down. Pad IDs follow Arduino's V0.3 schematic, while printed D/A labels follow its official top-view pinout. VBUS is called VUSB in that pinout. Every printed header label remains visible; unused outside callouts are omitted. Connected pins show both the board label and ESP32 GPIO. GPIO numbers match `src/board.hpp`, independently of Arduino pin-numbering preferences.

U2 is Waveshare 24382. Its external P2 connector stays in physical order: 1 BL, 2 RST, 3 D/C, 4 CS, 5 SCL/CLK, 6 SDA/DIN, 7 GND, 8 VCC. SCL/SDA are SPI signals. BL and VCC connect to Nano 3V3. Left/Right enclosure selection changes screen rotation and visual switch order, not wiring or physical tap identity.

Nano A0/GPIO1 reads the ladder; A1/GPIO2 and A3/GPIO4 read pedal TIPs. Each RING receives 3V3 through 5.1k (R20/R30), each TIP receives its bias output through 10k (R21/R31). Bias is D2/GPIO5 or D3/GPIO6: HIGH in resistor mode, high impedance in pot/disabled/boot states, never LOW. Pedals are passive 0–10k variable resistors (TIP–SLEEVE) or pots (outer legs RING/SLEEVE, wiper TIP). ADC voltages are used directly without a gain multiplier. Capture actual endpoints after assembly.

J2 is Amphenol ACJS-MHDC: manufacturer TOP is pedal 1 (10 SLEEVE, 11 RING, 12 TIP), BOTTOM is pedal 2 (4 SLEEVE, 5 RING, 6 TIP). Normal contacts stay unconnected; pad 13 is mechanical. SW11 uses A/B/click on D4/GPIO7, D5/GPIO8, D6/GPIO9 with internal pull-ups and common GND. Its pin labels are functional; verify terminal order on the purchased encoder.

The optional Pololu 2535 chain contains only LEDs and wires. D7/GPIO10 goes directly to LED1 DIN; each DOUT goes to the next DIN. Final DOUT stays open. Supply every LED from regulated 5V and common GND, rather than Nano 3V3. Physical side view: lens left, leads right, flat side down; top to bottom DIN, 5V, GND, DOUT. The manufacturer photograph is included unchanged as orientation reference. Verify 3.3V data acceptance with the actual batch.

Populate four ladder stages for the base; each extension adds two 1k resistors and switches. Select 4/6/8/10 in software and calibrate the installed taps. No end termination is fitted. Fixed tap/LED identity: T1=B, T2=A, T3=2, T4=1, T5=C, T6=3, T7=D, T8=4, T9=E, T10=5.

J1 accepts 9V DC center-negative: center GND, sleeve +9V. +9V feeds both Nano VIN (JP1:15) and Pololu S13V10F5 VIN. C1 is 47uF/35V across converter VIN/GND. Converter VOUT supplies the optional 5V LED rail only. Never join it to USB VBUS. Nano VBUS (JP1:12) feeds R50 100k, then A4/GPIO11; R51 150k connects that node to GND. This divider is required for self-powered USB detection. USB-C carries native USB MIDI and ROM flashing; GPIO19/20 are reserved internally for USB. For flashing, join B1 to GND, tap RST, remove the jumper, then select the ROM serial port.

`single-page-wiring.json` records terminals and wire geometry. The generator checks full net coverage, continuous routes, body crossings, and foreign-net overlaps. Dots indicate junctions; arched crossings remain separate. Both pedal ports share one socket outline. Reference numbering has deliberate gaps. Rebuild with `scripts/preview_electrical.py`; connectivity, routing and firmware-pin agreement are checked during generation. Manufacturer links are in `netlist.json`. Physical tolerances, noise, electrical protection and simultaneous-supply performance require hardware tests.